

Design and Prototyping of Computational Sunglasses for Autism Spectrum Disorders

Xiaodan Hu*
NAIST, Japan

Yan Zhang†
NAIST, Japan

Naoya Isoyama‡
NAIST, Japan

Nobuchika Sakata§
NAIST, Japan

Kiyoshi Kiyokawa¶
NAIST, Japan

ABSTRACT

We design the computational sunglasses for individuals with autism spectrum disorders (ASDs) to alleviate the atypical visual perceptions caused by their unique light sensitivities. With the proposed system, a scene camera is used for ambient illuminance detection, and an appropriate occlusion pattern is calculated in real-time then displayed on a spatial light modulator (SLM) to modulate the contrast of the incoming scene. Based on a simple contrast adjustment algorithm that avoids overexposure and underexposure, our bench-top prototype demonstrates better scene contrast successfully.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Computing methodologies—Artificial intelligence—Computer vision—Computer vision problems

1 INTRODUCTION

Autism spectrum disorders (ASDs) consist of a group of developmental disabilities characterized by atypical development in socialization, communication, and behaviors. Atypical visual perception and integration have been recognized as key phenotypes in individuals with ASD. Fan et al. proposed that individuals with ASD show significantly longer pupillary light reflex latency, smaller constriction amplitude, and lower constriction velocity than individuals with typical development [1]. More recently, Nagai et al. introduced several patterns of the atypical visual perception, including high-contrast or extremely bright vision stimulated by ambient illuminance and monochromatic blurry vision stimulated by dynamic objects, which suggests that individuals with ASD suffer from their atypical light sensitivity [2].

To mitigate the inconvenience caused by this issue, some of them wear sunglasses to darken the scene uniformly to prevent such stimulation. However, some information will be lost in the less-illuminated vision, which brings another inconvenience. Occlusion-capable augmented reality (AR) displays [4] are developed in recent years that are capable of precisely darkening bright areas in user's view, however, the high cost and bulkiness prevent them from daily usage. Nayar et al. proposed a system with a spatial light modulator (SLM) and an adaptive dynamic camera to modulate the scene brightness [3]. Motivated by the research, we design the computational sunglasses. The contrast of the observed scene is modulated by a single transmissive liquid crystal display (LCD). An additional high dynamic range (HDR) camera is used for capturing the front scene and calculating the occlusion pattern. The proposed system is expected to help individuals with ASD in their daily lives.

*e-mail: hu.xiaodan.ht1@is.naist.jp

†e-mail: zhang.yan.zq6@is.naist.jp

‡e-mail: isoyama@is.naist.jp

§e-mail: sakata@is.naist.jp

¶e-mail: kiyo@is.naist.jp

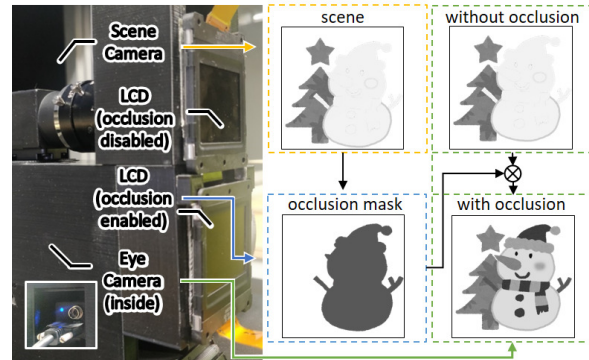


Figure 1: (Left) The prototype consists of two transmissive LCDs, a scene camera, and an eye camera that simulates the user's eye. (Right) The process flow is shown. The real environment is first captured by the SC (top left) through the secondary LCD with maximum transmittance. The original image of the EC with maximum transmittance (top right) is almost identical to the SC image with slight parallax. Then, an appropriate occlusion mask is computed and displayed on the primary LCD for the EC (bottom left) so that the EC captures images with more balanced contrast (bottom right).

2 SYSTEM OVERVIEW

Our computational sunglasses for individuals with atypical light sensitivity are composed of two parts; a light modulation part and an illuminance detection part (Fig. 1(left), Fig. 2(a)). The former consists of a primary LCD which displays an occlusion pattern for a human eye, and the latter consists of a scene camera (SC) and a secondary LCD with uniform transmittance that are positioned just above the primary LCD to minimize the parallax. The secondary LCD guarantees that the SC captures similar images as the human eye through the primary LCD without pattern dimming. In the prototype system we built, an eye-simulating camera (EC) is placed at the position of the human eye. A simple direct occlusion approach by a transmissive LCD is used because precise hard-edge occlusion requires a bulky device [4] and occlusion leaking is acceptable to some extent for the purpose of naturally attenuating the scene.

Figure 1(right) shows the process flow with pictures illustrating images taken by the SC and the EC. The real environment is first captured by the SC (top left) through the secondary LCD with maximum transmittance. The original image of the EC with maximum transmittance (top right) is almost identical to the SC image with slight parallax. Then, an appropriate occlusion mask is computed and displayed on the primary LCD for the EC (bottom left) so that the EC captures images with more balanced contrast (bottom right).

There are various possible strategies for computing the occlusion mask. Some of reasonable requirements include 1) high-intensity regions are dimmed and discernible for comfortable viewing, 2) low-intensity regions are not dimmed and kept nearly intact, and 3) regions with higher (lower) intensity appear brighter (darker) after modulation. We describe a simple algorithm used for the prototype system in Sec. 3.2 that satisfies these requirements.

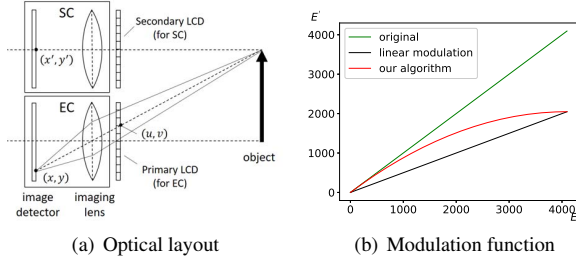


Figure 2: (a) Optical layout of the prototype. (b) Modulation functions between original (E) and target (E') pixel intensities; no modulation (green), linear modulation (black), and parabolic modulation (red).

3 PROTOTYPE SYSTEM

3.1 Apparatus and Calibration

The prototype system was developed by using two identical monochrome industrial HDR cameras (12bit depth) with a SONY IMX 287 sensor (720×540 pixels) and a lens with focal length of 6mm, $f/1.6$. The aperture is approximately equal to the large baseline pupil diameter. We used two sets of a SONY LCX017 panel and a pasted linear polarizer as the primary and the secondary LCDs. The calibration between the SC and the EC and between the EC and the primary LCD are done by finding two homography matrices by manually assuming a target distance and specifying corresponding feature points.

3.2 Occlusion Algorithm

Figure 2 describes the modulation functions, where x-axis (E) and y-axis (E') represent pixel values observed by the EC before and after modulation, respectively. Notice that E'/E gives transmittance T for given E . To satisfy the three requirements in Sec. 2 for computing an appropriate occlusion mask, we simply used a parabolic curve between the no-modulation function and a linear modulation function (uniform dimming), which has the following properties; 1) it is tangent to the no-modulation function at the lowest intensity, 2) it intersects with the linear-modulation function at the highest intensity, and 3) it increases monotonically. The slope of the linear modulation can be adjusted considering the light sensitivity of the individual. In the experiment, we simply set $T = 0.5$.

To quantify the transmittance T with certain pixel value v on the LCX017 whose gray scale depth is 8-bit, we put a luminance meter after the LCD to measure the incident light intensity. The relationship can be fitted by a sigmoid function

$$T = \frac{1}{1 + e^{-k(v-v_0)}} \quad (1)$$

where v_0 is a base pixel value and k is a coefficient of the sigmoid function. Thus the appropriate pixel value of corresponding points on the occlusion mask can be easily calculated by the reverse function of equation (1).

3.3 Experiment

Figure 3 shows example images of an indoor scene taken by the EC with different modulation functions in Fig. 2(b). The snowman on the table is overexposed without modulation (Fig. 3(a)). With linear modulation (uniform dimming), it is more visible and discernible (Fig. 3(b)), however, the darkly illuminated region on the lower left including a dogū statue is now underexposed and not visible. With our algorithm, the overexposed snowman and the darkly illuminated dogū are both visible (Fig. 3(c)) modulated with the occlusion mask in Fig. 3(d), satisfying the three requirements mentioned in Sec. 2.

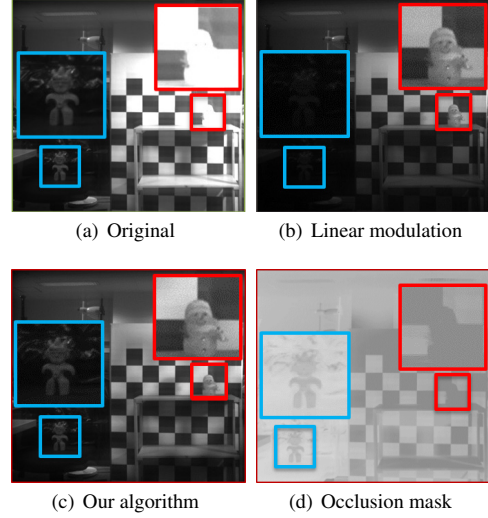


Figure 3: (a) to (c) Comparison of three observed images of the EC modulated with three modulation functions in Fig. 2(b). (d) The occlusion mask corresponding to (c) computed by our algorithm.

4 CONCLUSION

In this article, we proposed the computational sunglasses for individuals with ASDs by dimming too bright regions in the view adaptively with a single LCD. A proof-of-concept prototype demonstrated perceptually more balanced images without overexposure compared with original images without modulation, and underexposure compared with uniform dimming.

This is just the beginning of the project and we have a list of issues we would like to address in the future. First of all, we would like to improve the quality of the modulated image. The occlusion mask imprecisely dim the designated areas since the camera aperture (also the human pupil if used in practice) is not an ideal pinhole. Out-of-focus pixels on the occlusion mask yield blurred dimming called occlusion leaking. In order to mitigate the problem with a single LCD, fitting the point spread function by convolution with a Gaussian should be considered [3]. We would also like to manufacture a wearable stereo version of the computational sunglasses. Downsizing is one issue, but a more important issue is automatic calibration between the user's eyes and the SC. To achieve this, we will employ an eye tracker to measure the position and size of each pupil, calculate the homography and a blur-compensated occlusion mask in real-time. Motion and scene prediction is of our interest as well, because the visual experience will be deteriorated with the overall motion-to-photon latency. Finally, we would like to evaluate the feasibility of our system with individuals with typical and atypical light sensitivity in various lighting environments.

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